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Employment

Software & DevOps Engineer	Backend Ace, NY, USA	Nov 2024 – Present
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- Database - Imported 14,536 restaurants for client (Cual.co), including their photos and reviews from Google Maps ensuring clean data with no duplicates and proper formatting using *Python*
- DevOps - Migrated server and DNS from DigitalOcean to *AWS*, streamlining deployment using Docker and Jenkins on AWS EC2 instances
- Development - Implemented backend and frontend adjustments to support the new data and functionality, using *PHP and Laravel for backend development and Next.js (React) for frontend integration*.
- Developed full stack medical website for clients like The Bronx Clinic, NY, USA.

Web & App Developer	Freelance	Jan 2023 – Present
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- Worked on various projects as a freelance developer focusing on Cross Platform Web and Mobile Apps
- Collaborated with clients to understand their requirements and providing high-quality tailored solutions
- Built ML-based Applications like Activity Recognition using LSTM, OpenCV Object Detection, IoT Intrusion Detection etc. using *React Native, Flutter and Python*

Associate Web Developer	Wizzartech, TO, CA	Apr 2022 – Dec 2022
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- Led Jr. Developers to finishing projects using Wordpress and JS keeping codebase in check with git
- Developed a Facebook Ad automation tool using selenium with python to post ads automatically
- Completed projects according to client requirements e.g. Forhad Realty, Momin Auto Sales

Chief Tech Expert & RA	Robotics & Automation Research Lab (RARL), JX, CN	Aug 2018 – Jan 2022
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- Completed several applications which ranked in national level competitions and achieved software copyright
- Developed Several Full Stack Applications - Self-lecture Attendance Service [Ranked], Pandemic-driven Exam Assistant (PEA) [Ranked], Raspberry Pi SCADA System for Agricultural Plant Monitoring

Education

Kanpur, UP, India	Indian Institute of Technology Kanpur	Fall 2022 – Fall 2024
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- M.Tech. in Computer Science and Engineering, August 2024. CPI: 7.29/10
- Graduate Coursework: Embedded and Cyberphysical Systems, Deep Learning for Computer Vision, Introduction to ML, Sensing Communication & Networking

Ganzhou, JX, China	Jiangxi University of Science & Technology	Mar 2018 – Jan 2022
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- B.Eng. in Computer Science and Technology, January 2022. Percentage: 91.01%
- Undergraduate Coursework: Data Structures & Algorithms, Operating Systems, Embedded Systems, Computer Networks, Database Management System

Technical Experience

Projects

- **RADAR+Camera Monitoring App** (Aug 2023 - Feb 2024). An app to interact with Radar OPS-243C with camera integration with speed and distance detection of objects. Python, PyQt5
- **IIT Kanpur Firewall Authenticator GUI** (Sep 2023 - Aug 2023). A GUI wrapper for auto-authentication of IIT Kanpur's Firewall with keep alive and background execution for embedded devices. Python, PyQt5

- **Activity Recognition Application using ML** (Sep 2023 - Aug 2023). Mobile application to live monitor activities using motion sensors and ML Model in cloud. Flutter (Dart), Python
- **Smart Parking System** (Jul 2022 - Sep 2022). Web application to connect to cloud and LoRaWAN based parking modules to manage multiple parking lots. ReactJS, ExpressJS
- **Car Dealership CRM** (May 2022 - Jun 2022). A full stack CRM application to get leads, record customer information, send promotional emails, etc. ReactJS, Material UI, ExpressJS
- **Online Exam Assistant** (Oct 2020 - Apr 2021). A full stack android application for distance education having course management, exam scheduling, and timed assessment with anti-cheating mechanisms. Kotlin, PHP
Ranked 2nd Position (Provincial) & 3rd Position (National) - 14th China University Computer Design Competition.
- **Self-Lecture Attendance System** (May 2019 - Dec 2020). A full stack QR-based attendance system for WeChat to securely give attendance at institutions. WeChat Mini Program (JavaScript), Java Springboot
Won 2nd Prize in Central China - "Internet " Innovation Competition and the 7th "Discovery Cup", China

Academic Experience

Projects

- **IoT based on-site Water Pipeline Inspection (Summer 2023):** Implemented efficient channel attention mechanism inside YOLOv7 object detection model and reduced model complexity to increase accuracy
- **Robot Assistant X - Your Personal ChatBot (Fall 2023):** Implemented an NLP Model using Tensorflow for the Raspberry Pi based robot and built a Companion App for the Robot using Flutter
- **Automated Guided Vehicle (AGV) Path Tracking (Fall 2022):** Designed and simulated Pioneer 3DX robot's path following across traffic intersection with timed automata in Coppeliassim VREP
- **Remote Traffic Police Sensing via Deep Learning (Mar 2021 - Dec 2022):** Implemented a multi-label object detection model for moving vehicles using OpenCV and deployed speed detection algorithm using OpenVINO on Raspberry Pi accounting for camera tilt angles

Awards and Certifications

- 2023 - **MHRD Fellowship** (IIT Kanpur)
- 2022 - **Outstanding International Graduate Award** – JXUST (Undergraduate)
- 2022 - **Software Copyright CN** - Pandemic-driven Exam Assistant (PEA). Copyright No: 2022SR0088310
- 2021 - **3rd Position in National Level** - 14th China University Computer Design Competition
- 2020 - **Software Copyright CN** - Self-Lecture Attendance System (SLAS) Copyright No. 2020SR0058411
- 2019 - **Google IT Support Specialization** (Issued: May 2019) – Credential ID: XWWFL3BT84WV
- 2019 & 2020 - **Jiangxi Provincial Government Scholarship** for foreign students, China
- 2019 - **2nd Prize in Central China** - "Internet " Innovation Competition and the 7th "Discovery Cup", China

Languages and Technologies

- C/C++, C# (Unity), Python, Javascript (NodeJS/ReactJS), HTML/CSS, SQL, PHP (Laravel), JAVA/Kotlin (Android), SpringBoot, Flutter (Dart)
- Scikit-learn, Pandas, Numpy, PyTorch, NoSQL (MongoDB), PyQt5, Git
- AWS, Digital Ocean, Google Cloud Platform, Docker, Jenkins